

Spring 2025 MC Problem 3: Perfect Complements Marshallian vs. Hicksian Demand

ECO 310

1 The Problem

Assume utility $u(x_1, x_2) = \min\{4x_1, x_2\}$. The Hicksian demand bundle is given by:

- (a) $(4u^*, u^*)$
- (b) $(\frac{u^*}{4}, u^*)$ ✓ CORRECT
- (c) $(u^*, \frac{u^*}{4})$
- (d) $(u^*, 4u^*)$

The problem asks for **Hicksian demand**, but let's compare it with Marshallian demand to understand the difference.

2 Perfect Complements Intuition

With $u(x_1, x_2) = \min\{4x_1, x_2\}$, goods are consumed in a **fixed ratio** to avoid waste.

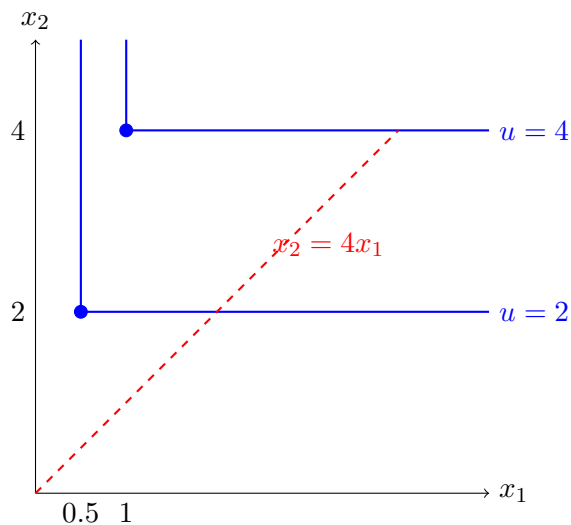
2.1 The Kink Condition

Utility is determined by the minimum of $4x_1$ and x_2 . To maximize utility without wasting money:

$$\boxed{4x_1 = x_2}$$

This is like left shoes and right shoes: you need them in a specific ratio. Here, for every 1 unit of good 1, you need 4 units of good 2.

2.2 Graphical Representation



The consumer always chooses bundles along the line $x_2 = 4x_1$ (the red dashed line where the kinks occur).

3 Marshallian Demand (Income Fixed)

3.1 The Problem

Maximize utility subject to a budget constraint:

$$\max_{x_1, x_2} \min\{4x_1, x_2\} \quad \text{subject to} \quad p_1x_1 + p_2x_2 = m$$

What's fixed: Income m and prices (p_1, p_2)

Question: "Given my income m , what should I buy?"

3.2 Solution

Step 1: Set the kink condition $4x_1 = x_2$ (optimal allocation)

Step 2: Substitute into the budget constraint:

$$p_1x_1 + p_2x_2 = m$$

$$p_1x_1 + p_2(4x_1) = m$$

$$x_1(p_1 + 4p_2) = m$$

Step 3: Solve for x_1 and x_2 :

$$x_1^*(p_1, p_2, m) = \frac{m}{p_1 + 4p_2}$$

$$x_2^*(p_1, p_2, m) = 4x_1^* = \frac{4m}{p_1 + 4p_2}$$

3.3 Properties

- Depends on **income** m and **prices**
- Observable in real markets
- Also called “uncompensated demand”

4 Hicksian Demand (Utility Fixed)

4.1 The Problem

Minimize expenditure subject to achieving a target utility:

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \quad \text{subject to} \quad \min\{4x_1, x_2\} = u^*$$

What’s fixed: Target utility u^* and prices (p_1, p_2)

Question: “What’s the cheapest way to achieve utility u^* ?”

4.2 Solution

Step 1: Set the kink condition $4x_1 = x_2$ (optimal allocation)

Step 2: Substitute into the utility constraint:

$$u^* = \min\{4x_1, x_2\} = \min\{4x_1, 4x_1\} = 4x_1$$

Step 3: Solve for x_1 :

$$x_1 = \frac{u^*}{4}$$

Step 4: Use the kink condition to find x_2 :

$$x_2 = 4x_1 = 4 \cdot \frac{u^*}{4} = u^*$$

4.3 Final Answer

$$h_1(p_1, p_2, u^*) = \frac{u^*}{4}$$

$$h_2(p_1, p_2, u^*) = u^*$$

This is answer **(b)**: $(\frac{u^*}{4}, u^*)$

4.4 Properties

- Depends on **target utility** u^* (not income!)
- For perfect complements, **independent of prices** (the ratio is always fixed)
- Also called “compensated demand”
- Theoretical construct, not directly observable

5 Key Difference: Prices Don't Affect Hicksian for Perfect Complements

Notice something surprising: Hicksian demand $(\frac{u^*}{4}, u^*)$ doesn't depend on p_1 or p_2 !

Why? With perfect complements, you **must** consume in the fixed ratio $x_2 = 4x_1$ no matter what. If you want utility u^* , you need exactly:

- $\frac{u^*}{4}$ units of good 1
- u^* units of good 2

There's no substitution possible. Prices only affect the **total cost**, not the **quantities**.

6 Comparison Table

Feature	Marshallian	Hicksian
Problem	Max u s.t. $p \cdot x = m$	Min $p \cdot x$ s.t. $u = u^*$
Fixed variable	Income m	Utility u^*
Kink condition	$x_2 = 4x_1$	$x_2 = 4x_1$
x_1 demand	$\frac{m}{p_1 + 4p_2}$	$\frac{u^*}{4}$
x_2 demand	$\frac{4m}{p_1 + 4p_2}$	u^*
Depends on prices?	Yes	No (for perfect compl.)
Depends on income?	Yes	No
Depends on utility?	No (achieved, not fixed)	Yes (fixed)

7 Numerical Example

Let $p_1 = 1$, $p_2 = 2$, and $m = 18$.

7.1 Marshallian Approach

Find the optimal bundle given income $m = 18$:

$$x_1^* = \frac{18}{1 + 4(2)} = \frac{18}{9} = 2$$

$$x_2^* = \frac{4(18)}{1 + 4(2)} = \frac{72}{9} = 8$$

Check: $p_1x_1 + p_2x_2 = 1(2) + 2(8) = 18$ ✓

Achieved utility: $u^* = \min\{4(2), 8\} = \min\{8, 8\} = 8$

7.2 Hicksian Approach

Now suppose we want to achieve utility $u^* = 8$ as cheaply as possible:

$$h_1 = \frac{8}{4} = 2$$

$$h_2 = 8$$

Total expenditure: $e(p_1, p_2, u^*) = 1(2) + 2(8) = 18$

7.3 Duality

Notice: Both approaches give the **same bundle** (2, 8)!

This demonstrates the duality relationship:

$$x^*(p, m) = h(p, u^*) \quad \text{where } u^* = u(x^*(p, m))$$

The Marshallian demand at income m equals the Hicksian demand at the utility level achieved with income m .

8 Why Answer (b) is Correct

The question asks for Hicksian demand with utility u^* . Using the constraint $\min\{4x_1, x_2\} = u^*$ and the kink condition $4x_1 = x_2$:

$$\begin{aligned} u^* &= 4x_1 = x_2 \\ \Rightarrow x_1 &= \frac{u^*}{4} \\ \Rightarrow x_2 &= u^* \end{aligned}$$

Therefore, the Hicksian demand bundle is: $\boxed{\left(\frac{u^*}{4}, u^*\right)}$ which is answer **(b)**.

9 Common Mistakes

1. **Confusing Marshallian with Hicksian:** Marshallian depends on m , Hicksian depends on u^*
2. **Getting the ratio backwards:** The constraint is $u = \min\{4x_1, x_2\}$, so we need $4x_1 = x_2$, which means $x_1 = \frac{x_2}{4}$ or equivalently $x_1 = \frac{u^*}{4}$
3. **Forgetting prices don't matter:** For perfect complements, Hicksian demand doesn't depend on prices because the ratio is fixed

10 Summary

- **Perfect complements** require fixed consumption ratio: $x_2 = 4x_1$
- **Marshallian demand:** Given income m , buy $\left(\frac{m}{p_1 + 4p_2}, \frac{4m}{p_1 + 4p_2}\right)$
- **Hicksian demand:** To achieve utility u^* , buy $\left(\frac{u^*}{4}, u^*\right)$
- For perfect complements, Hicksian demand is **independent of prices**
- The two demands are related through duality: same bundle at corresponding income/utility levels